The Winter Bottleneck Progress report

Project scope update

The project scope is narrowed to specifically using the greater Boston area weather as an indicator of weather data, instead of the planned coordinate weather details as the retrieval of such API data remains a difficulty at the current stage of the project.
At the current stage, the project has completed the foundational data retrieval and preprocessing pipeline. The code can now automatically download Bluebikes trip and station data, retrieve historical hourly weather data, clean and standardize the datasets, and construct a preliminary merged dataset for analysis. The next phase of the project will focus on refining the analytical logic and producing summary outputs and visual analysis.

Data sources

So far, three data sources have been retrieved using the python code i've written.

1. Bluebike trip history data of November 2025: downloaded from the bluebike system data archive. The code automates the downloading process and also unzipping process of the code to a csv. This dataset includes trip level information such as ride start and end time, start and end station and station IDs, coordinates of the station, and rider type. The current file contains 344,841 rows and 13 columns.
2. Blue station list of November 2025: downloaded as a csv file from the bluebike system data page. This file includes station data such as station number, station name, latitude, longitude, municipality, total docks, and station ID. The current file contain 597 rows and 8 columns
3. Historical hourly weather data: downloaded using the Open-Meteo weather API. Data ranges from the period of October 31, 2025 to November 30, 2025. The data include variables such as temperature, precipitation, snowfall, windspeed, cloud cover. This weather data currently contain 744 hourly observations and 6 columns.

Issues / Difficulties

The first and also the biggest challenge that I've met so far is regarding the weather API integration. Initially, I thought I could use the open-meteo API's bounding box feature to get a full scale weather data of a series of grids, then match these grids with the station coordinates to differentiate between the weather conditions of each station for more sophisticated and refined analysis. However, this approach appears to be significantly more complex than I've expected. As a result, I've managed to simplify the weather data retrieval to using a generalized coordinate of the greater boston area and first managed to establish a stable API workflow using that simplified coordinate. A future challenge will be deciding whether to continue with this representative weather approach or use the more refined method and specify the data to match precisely to station coordinates. Considering there are 597 stations, each with different coordinates, and that I'm calculating the span of an entire month with hourly data update, the feasibility of using coordinate specific weather data remain undecided.